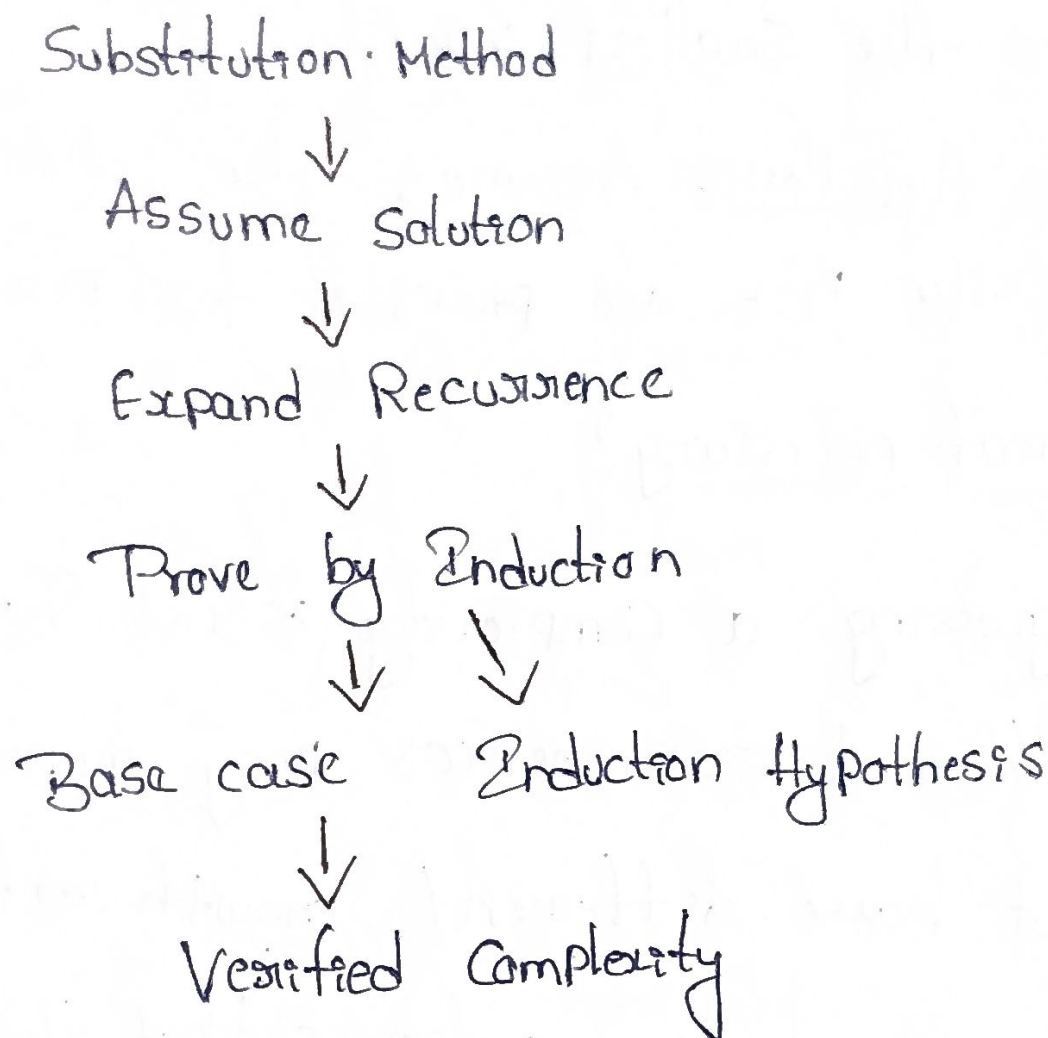


③ Substitution Method - Detailed Explanation

Extended Concept Map:



Explanation:-

The Substitution Method is used to solve recurrence relations by first assuming a possible time complexity. The recurrence ~~A~~ is then expanded algebraically to check whether the assumed expression fits the recurrence. After expansion.

the Solution is proved Using mathematical induction, which consists of:

- Base case: Verifies that the assumed solution is true for the smallest input.

- Induction Hypothesis: Assumes the Solution is true for the $n=k$ and proves it for $n=k+1$.

Why is proof necessary?

Simply guessing a Complexity is not Sufficient because different recurrences may appear

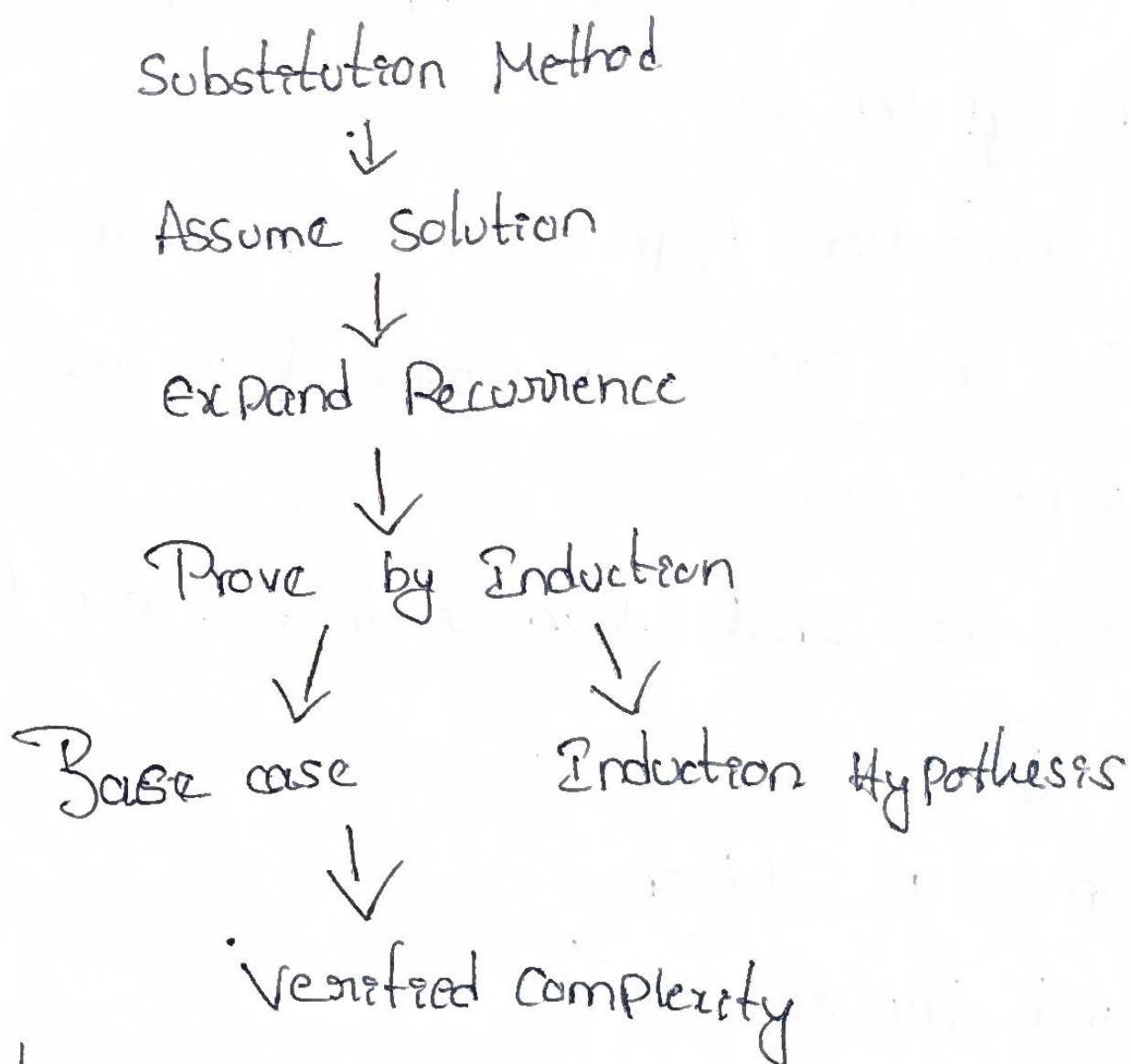
Similar but have different growth rates.

Mathematical induction confirms that the assumed Solution Satisfies the recurrence for all values of n , ensuring correctness and eliminating incorrect assumptions.

Solution is proved using

⑦ Substitution Method - Simple Answer

Extended Concept Map



Explanation:-

The Substitution Method is Used to solve recurrence relations by first assuming a time complexity. The recurrence is then expanded step by step to identify a pattern. Finally, the assumed solution is proved using mathematical induction.

to verify that it is correct.

Base Case:-

The Base case checks whether the assumed solution is true for the smallest input (usually $n=1$)

Induction Hypothesis:-

The Induction Hypothesis assumes the solution is true for one case and proves it is also true for the next case.

How does the substitution method derive the time complexity?

1. Assume a solution.
2. Expand the recurrence
3. Identify the pattern.
4. Prove the solution using induction.
5. Get the verified time complexity.